

Rick S.H. Willemsen

Singapore University of Technology and Design
Engineering Systems and Design

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Date of birth: 24-01-1997

Research interests

Methods: mathematical programming, decomposition techniques

Applications: clustering, dimension reduction, data generation, transportation, brain connectomes, cancer identification

Academic employment

Singapore University of Technology and Design , Singapore	Jan 2026 – Jan 2028
Postdoctoral research fellow	
PI: Karthik Natarajan	
Erasmus University Rotterdam , The Netherlands	May 2025 – Aug 2025
Researcher	

Education

Erasmus University Rotterdam , The Netherlands	Sept 2020 – Apr 2025
PhD in Operations Research and Statistics	
Dissertation: Mathematical programming in dimension reduction and clustering	
Advisors: Wilco van den Heuvel and Michel van de Velden	
Polytechnique Montréal , Canada	Sept 2023 – Nov 2023
Research visit	
Hosts: Daniel Aloise and Raf Jans	
Erasmus University Rotterdam , The Netherlands	Sept 2019 – Mar 2021
MSc in Econometrics and Management Science	
Specialisation: Business Analytics and Quantitative Marketing	
Erasmus University Rotterdam , The Netherlands	Sept 2019 – Aug 2020
MSc in Econometrics and Management Science	
Specialisation: Operations Research and Quantitative Logistics	
Erasmus University Rotterdam , The Netherlands	Sept 2015 – Aug 2019
BSc in Econometrics and Operations Research	
BSc in Economics and Business Economics	
Minor in Advanced Computer Science	
City University , Hong Kong	Sept 2018 – Dec 2018
Exchange semester	

Research papers & publications

Journal articles

Willemsen R. S. H., Abeywickrama T. & Anandakrishnan R. (2026). [Identifying multi-hit cancer drivers without massive parallelization: a CP, MIP, and column generation framework](#). *International Conference on Principles and Practice of Constraint Programming*.

- Willemsen R. S. H., Cavicchia C., Van den Heuvel W. & Van de Velden M. (2026). [An exact solution approach for hierarchical clustering](#). *INFORMS Journal on Computing*, 382, pp. 447–462.
- Willemsen R. S. H., Van de Velden M. & Van den Heuvel W. (2024). [On the uniqueness of correspondence analysis solutions](#). *Linear Algebra and its Applications*, 690, pp. 162–185.
- Willemsen R. S. H., Van den Heuvel W. & Van de Velden M. (2023). [A new mixed integer programming approach for inverse correspondence analysis](#). *Computers & Operations Research*, 160, p. 106375.

Submitted

- Burer S., Natarajan K. & Willemsen R. S. H. (2026). [On the semidefinite representability of continuous quadratic submodular minimization with applications to moment problems](#). *preprint available, major revision at Operations Research*.
- De Kort M. A. A., Willemsen R. S. H. & Van den Heuvel W. (2026). A column generation approach for interpretable and fair clustering. *submitted to Omega*.

Working papers

- Bouman P. C., Tiniç G. Ö., Willemsen R. S. H. & Veelenturf L. P. (2026). From self driving cars to the next generation of public transport.
- Willemsen R. S. H., Ho T. L. F. & Abeywickrama T. (2026). Exact and efficient methods for identifying carcinogenic multi-hit gene combinations.
- Willemsen R. S. H. (2026). A Benders decomposition approach for k -medoids clustering.
- Willemsen R. S. H., Aloise D. & Jans R. (2025). [Accelerating Benders decomposition for the \$p\$ -median problem through variable aggregation](#). *preprint available*.
- Willemsen R. S. H., Van de Velden M. & Van den Heuvel W. (2026). Generating contingency tables with given marginals and levels of association.
- Willemsen R. S. H., Van den Heuvel W. & Van de Velden M. (2025). [Generating vectors with linear and nonlinear constraints](#). *preprint available*.

Research experience

- Graduate intern**, Netherlands Railways, Utrecht, The Netherlands Apr 2020 – Aug 2020
 Topic: a column generation approach for the integrated crew rostering problem with fairness and attractiveness requirements
 Advisor: Dennis Huisman
- Research assistant**, Erasmus University Rotterdam, The Netherlands Feb 2019 – Aug 2020
 Topic: the pickup and delivery problem with online transfers, for the next generation of public transport
 Advisor: Paul Bouman
- Research assistant**, Erasmus University Rotterdam, The Netherlands Feb 2018 – Aug 2018
 Topic: optimal interventions in network models
 Advisor: Evelot Duijzer

Work experience

- Intern**, Pointlogic, A Nielsen Company, Rotterdam, The Netherlands Feb 2018 – Aug 2018

Teaching experience

- Teaching assistant**, Erasmus University Rotterdam, The Netherlands Sept 2016 – Aug 2025
 Teaching tutorial groups (25 students) and exercise lectures (>100 students) on bachelor level. Creating and grading assignments and exams. Supervising bachelor and master theses.
- Introduction to Statistics
 - Introduction to Analysis

Analysis
Introduction to Programming
Programming
Matrix Algebra
Introductory Seminar Case Studies Econometrics and Operations Research
Econometrics 1
Econometrics 2
Non-linear Optimisation
Academic Skills
Mathematical Methods
Bachelor Thesis supervisor
Master Thesis supervisor

Seminars & conference presentations

Peer-reviewed conference presentations

International Conference on Principles and Practice of Constraint Programming (CP), 2026, Lisbon, Portugal
Pitch presenter, 12th Heidelberg Laureate Forum (HLF), 2025, Heidelberg, Germany
12th Triennial Symposium on Transportation Analysis (TRISTAN), 2025, Okinawa, Japan

Invited session speaker

INFORMS International Meeting, 2025, Singapore
European Conference on Data Analysis (ECDA), 2022, Naples, Italy

Invited seminars

Operations Research Seminar, 2026, Erasmus University Rotterdam, The Netherlands
EURO Online Seminar Series on Operational Research and Machine Learning, 2025, Virtual
Business Analytics Seminar, 2024, University of Amsterdam, The Netherlands

Contributed conference presentations

IFORS, 2026, Vienna, Austria
INFORMS Annual Meeting, 2024, Seattle, United States
ISMP, 2024, Montréal, Canada
EURO, 2024, Copenhagen, Denmark
INFORMS Annual Meeting, 2023, Phoenix, United States
Column Generation Workshop, 2023, Montréal, Canada
LNMB (Dutch Network for Operations Research), 2023, Soesterberg, The Netherlands
Dutch/Flemish Classification Society (VOC), 2022, Tilburg, The Netherlands
EURO, 2022, Espoo, Finland
EURO, 2021, Athens, Greece
Joint conference on Data Science, Statistics & Visualisation and European Conference on Data Analysis (DSSV-ECDA), 2021, Virtual

Internal seminars & colloquia

Operations Research Seminar, 2025, Erasmus University Rotterdam, The Netherlands
Operations Research Seminar, 2024, Erasmus University Rotterdam, The Netherlands
GERAD Seminar 2023, Montréal, Canada
Econometric Institute Internal Seminar, 2023, Erasmus University Rotterdam, The Netherlands
Operations Research Seminar, 2022, Erasmus University Rotterdam, The Netherlands
Econometric Institute PhD Conference, 2022, Erasmus University Rotterdam, The Netherlands

Achievements

Global Young Scientists Summit 2026 , selected as one of ~400 young scientists worldwide to attend the GYSS2026 in Singapore	2026
12th Heidelberg Laureate Forum , selected as one of 200 young researchers worldwide to attend the HLF2025 in Heidelberg	2025
Princeton University's Maximum Quasi-Cliques Challenge , 3 rd place	2025
ROADEF2022 3D Truck Loading Competition , 1 st place in junior category	2023
FAECTOR Excellence Programme on Econometrics & Artificial Intelligence , selected participant	2018

Services

Reviewer: NeurIPS, Journal of Classification, Networks and Spatial Economics, International Transactions in Operational Research
Session chair: INFORMS2023, INFORMS2024

Skills

Programming languages: Java, Python, C++ , MATLAB, R, MySQL

Languages

Dutch: native
English: full professional proficiency
Chinese: limited working proficiency

References

Wilco van den Heuvel

Full Professor at the Econometric Institute, Erasmus School of Economics, Erasmus University Rotterdam
wvandenheuvel@ese.eur.nl

Michel van de Velden

Associate Professor at the Econometric Institute, Erasmus School of Economics, Erasmus University Rotterdam
vandevelden@ese.eur.nl

Daniel Aloise

Full Professor at the Department of Computer Engineering and Software Engineering, Polytechnique Montréal
daniel.aloise@polymtl.ca